

**CORRELATIONS OF ARTIN STATUS AMONG PRIMES:
CONSECUTIVE-PRIME ANTICORRELATION, CROSS-BASE
ENTANGLEMENT,
AND AN ELEMENTARY DECOMPOSITION**

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ABSTRACT. Say a prime p is *Artin base a* if a is a primitive root modulo p . This paper studies correlations of Artin status along two axes — *consecutive primes* at a fixed base, and *several bases* at a fixed prime — for all primes up to 10^9 (with 50,847,531 of them, $p \geq 7$, in the consecutive census), and then asks what, if anything, such correlations are computationally worth.

Along the first axis, Artin statuses of consecutive primes are measurably *anticorrelated* at base 10: $P(\text{Art}_{n+1} \mid \text{Art}_n) = 0.36514$ against $P(\text{Art}_{n+1} \mid \neg \text{Art}_n) = 0.37928$, a deficit $\delta = -0.01414$ (Pearson 10,167 on one degree of freedom; signed square root -100.8 under a nominal independent-status reference, over 50,847,530 pairs). The dependence has strong gap structure and contains one deterministic component, an elementary *exclusion law*: if $p, p+g > 5$ are prime with $g \equiv 20 \pmod{40}$, then 10 is a quadratic residue modulo exactly one of them, so the two can never both be Artin; among 2,195,882 pairs at gaps 20 and 60 there is no doubly-Artin pair, and an independent recomputation extends this to 194,296,748 such pairs below 10^{11} , again with none. Conversely $g \equiv 0 \pmod{40}$ forces equal quadratic-residue status and strong positive dependence ($P(\text{Art}_{n+1} \mid \text{Art}_n) = 0.899$ at $g = 40$). Conditioning on the joint residues of (p_n, p_{n+1}) modulo 120 or 840 places 98.7% and 99.5% of the global deficit between joint-residue cells rather than within them (an exact covariance decomposition; 96.1–99.2% and 97.8–99.7% across the four weighting conventions we report). As an intermediate link we report the apparently unreported anticorrelation $r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.041$.

Along the second axis we record a second exclusion law: writing $d = \text{sqf}(ab)$ for the squarefree part of ab , no prime with $\left(\frac{d}{p}\right) = -1$ is simultaneously Artin for a and for b ; whenever a and b have different squarefree parts this bars half of all primes, and it implies a triple form — if $\text{sqf}(c) = \text{sqf}(ab)$ then no odd prime is Artin for a, b, c at once (e.g. 2, 5, 10). Across the 66 base pairs of $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$ the measured same-prime correlation $\phi(a, b)$ is positive for 64 pairs (mean 0.0352); conditioning on $\omega(p - 1)$ removes only 55% of it, while conditioning on a fine signature of $p - 1$ removes 95%. The residual is concentrated on the fifteen pairs whose product class completes a triple and lies in the quadratic-character channel. Joint densities predicted by the Matthews–Moree–Stevenhagen Kummer-degree heuristic match the measured densities to a mean relative error of 0.012% with no fitted parameters.

Finally we *audit the computational content* of these correlations. In a resubstitution diagnostic the consecutive-prime correlation carries 0.000009 nats (1.3×10^{-5} bits) of conditional-entropy reduction beyond the elementary joint-residue baseline, which itself carries 0.4195 nats. A simulation in which every status depends only on $p \bmod 840$ reproduces the part of that increment which exceeds the mod-120 baseline: residues (the prime 7 of $p - 1$) suffice to reproduce that excess without predecessor-specific information, and in held-out scoring the predecessor’s status gives no gain beyond residues mod 840. Both exclusion laws are computationally redundant: the gap law from quadratic reciprocity together with multiplicativity, the same-prime law from multiplicativity alone. The quadratic filter expressed by these exclusion laws — a primitive root must be a quadratic non-residue — is elementary, is already applied by a character-aware implementation, is exactly what the observed conductor-40 channel expresses, and halves the factorizations of $p - 1$ in a scanning search; ordering the small prime channels of $p - 1$ before the full factorization is about 13% faster again. We conclude that, for the consecutive-prime workloads tested, these correlations describe arithmetic structure rather than supply algorithms, and we state what would change that verdict.

1. INTRODUCTION

Artin’s conjecture on primitive roots, communicated to Hasse in 1927 (see [1]), concerns one base at a time: for non-square $a \neq 0, \pm 1$ the set of primes p for which a is a primitive root mod p should have positive density, a multiple of Artin’s constant $C = \prod_q \left(1 - \frac{1}{q(q-1)}\right) \approx 0.3739558$; for base 10 no correction factor applies, so the conjectural density is exactly C . Hooley [11] proved this under GRH; unconditionally, Gupta–Murty [8] and Heath-Brown [10] showed the conjecture can fail for at most two prime bases, and Klurman–Shparlinski–Teräväinen [12]

established the predicted asymptotic on average over almost all small bases; recent work has sharpened the analytic understanding for individual primes, with Fan and Pollack [5] proving a version uniform in the base and Perucca and Shparlinski [16] giving uniform upper and lower bounds for the Artin density. Moree’s survey [15] is the standard account. For $a = 10$ the property has a classical meaning: 10 is a primitive root mod p exactly when the decimal expansion of $1/p$ has maximal period $p - 1$ — the *full reptend* primes [3, pp. 166–171].

Three further strands meet here. The first is the bias discovered numerically by Knapowski–Turán, Ko and Ash–Beltis–Gross–Sinnott and quantitatively treated by Lemke Oliver and Soundararajan [14]: consecutive primes in the same reduced residue class modulo q occur markedly less often than equidistribution predicts, a phenomenon explained conjecturally by the Hardy–Littlewood k -tuple conjectures [9]. Since the Artin property of p is governed by the factorisation of $p - 1$, and since residues of consecutive primes modulo small q are correlated by [14], one expects *some* induced dependence between the Artin indicators of consecutive primes. The second is the joint behaviour *across bases* at a fixed prime, computed under GRH by Matthews [24] and systematically by Moree–Stevenhagen [23], with the qualitative message that $\mathcal{A}_a$ and $\mathcal{A}_b$ are *not* independent and that the deviation is governed by entanglement of the associated Kummer–Galois extensions. The framework remains active: Lenstra’s Galois-theoretic setting [22], the unified Artin-type treatment of Järvinen–Perucca–Sgobba [20], unconditional results in cases where GRH can be circumvented [25], and number-field variants [21]. The third is the question of which primes at a fixed shift can *both* be Artin: Tinková–Waxman–Zindulka [13] identify shift–base pairs admitting no such prime, and our Theorem 2 is the base-10 instance of that phenomenon; we claim no priority for it.

The present paper asks the *statistical* question along both axes — how Artin statuses of consecutive primes are correlated on average, and how Artin statuses of several bases are correlated at the same prime — and then asks a question we have not seen posed in this literature: *what is such a correlation computationally worth?*¹ The answer we give is a measured negative, and we regard it as part of the result rather than a failure of it.

Two comparisons should be made explicit. Garcia–Kahoro–Luca [6] (see also [7]) compare the *number* of primitive roots, $\varphi(p - 1)$ versus $\varphi(p + 1)$, for twin pairs; Pollack [17] and Baker–Pollack [2] prove existence of runs of consecutive primes each having a prescribed primitive root. Those are existence results, produced by sieves that select rare favourable configurations; they say nothing about the typical joint behaviour along the sequence of primes. The questions studied here are complementary: the average correlation, its decomposition, and its computational content.

2. SUMMARY OF RESULTS

Throughout, *Artin prime* means Artin prime for base 10 unless stated otherwise, p_n denotes the n -th prime, $\omega(m)$ the number of distinct prime factors of m , and $\text{sqf}(m)$ the squarefree part of m .

- (1) **Consecutive-prime exclusion law** (Theorem 2). If p and $p + g$ are primes with $p, p + g > 5$ and $g \equiv 20 \pmod{40}$, then $\left(\frac{10}{p+g}\right) = -\left(\frac{10}{p}\right)$; consequently at most one of the two is Artin base 10. If instead $g \equiv 0 \pmod{40}$ then $\left(\frac{10}{p+g}\right) = \left(\frac{10}{p}\right)$.

¹An earlier version of parts of this work appeared as the preprints *Correlations between primitive root statuses of consecutive primes* (Zenodo, doi:10.5281/zenodo.22863946) and *Cross-base correlations of Artin primes* (Zenodo, doi:10.5281/zenodo.22878204). This paper consolidates those two preprints and adds the audit of §15; it supersedes them, and the preprints are retained for the record.

- (2) **Same-prime exclusion law** (Theorem 6). With $d = \text{sqf}(ab)$, no prime with $\left(\frac{d}{p}\right) = -1$ is Artin for both a and b . If $\text{sqf}(c) = \text{sqf}(ab)$ then no odd prime is Artin for a , b and c simultaneously (Theorem 7).
- (3) **Consecutive-prime anticorrelation** (§7). Across all 50,847,530 consecutive pairs up to 10^9 , $\delta = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n) = -0.01414$, with Pearson $\chi^2 = 10,167$ on one degree of freedom.
- (4) **Gap structure** (§8). $\delta(g)$ ranges from -0.592 at $g = 60$, where the theorem forces exclusion, to $+0.643$ at $g = 40$.
- (5) **Residue conditioning** (§9). Conditioning on the joint residues of (p_n, p_{n+1}) modulo 120 and 840 gives an exact covariance decomposition of the global deficit, placing 98.69% and 99.55% of it between cells (within-cell terms -0.000186 and -0.000064); across four weighting conventions the between share lies in $[96.1\%, 99.2\%]$ and $[97.8\%, 99.7\%]$.
- (6) **ω repulsion** (§10). $r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.0410$.
- (7) **Cross-base correlation** (§11). Over the 66 pairs of the base set, $\phi(a, b)$ is positive for 64 pairs (mean 0.0352); the only negative pairs are (2, 10) and (3, 15).
- (8) **Decomposition of the cross-base correlation** (§12). Conditioning on $\omega(p - 1)$ removes 55% of the mean signed correlation; conditioning on the fine signature $\text{sig}(p)$ removes 95% of it.
- (9) **Residual localisation** (§13). The surviving residual is concentrated on the triple-completing pairs and lies in the quadratic-character channel.
- (10) **Kummer-degree densities** (§14). The Matthews–Moree–Stevenhagen predictions match the measured joint densities to a mean relative error of 0.012%, with no fitted parameters.
- (11) **Computational content: a measured negative** (§15). The consecutive-prime correlation adds 0.000009 nats of in-sample conditional-entropy reduction to the mod-120 joint-residue baseline; a residue-only simulation mod 840 reproduces that excess without predecessor-specific information, and no held-out estimator we tested gains beyond residues mod 840; both exclusion laws are computationally redundant, the same-prime law from character multiplicativity alone and the gap law from reciprocity together with it; the quadratic filter expressed by these exclusion laws is the elementary non-residue test, which a character-aware implementation already applies and the conductor-40 channel expresses.

3. PRELIMINARIES AND NOTATION

For an odd prime p not dividing a , write $\left(\frac{a}{p}\right)$ for the Legendre symbol and $\chi_a(p) = \left(\frac{a}{p}\right)$ for the quadratic character of conductor $\text{cond}(\chi_a)$. Write $\mathcal{A}_a$ for the set of primes for which a is a primitive root, and $X_a(p) = \mathbf{1}[p \in \mathcal{A}_a]$ for its indicator; along the consecutive-prime axis we abbreviate $\text{Art}_n = \mathbf{1}[p_n \in \mathcal{A}_{10}]$ for Artin status at the n -th prime. Then a is a primitive root modulo p exactly when $a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p - 1$. Two elementary facts are used repeatedly. First, if $X_a(p) = 1$ then $\left(\frac{a}{p}\right) = -1$: a primitive root has order $p - 1$, so $a^{(p-1)/2} \equiv -1$, whereas a quadratic residue satisfies $a^{(p-1)/2} \equiv 1$. Second, quadratic characters are multiplicative in the numerator, $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$, and the Legendre symbol depends on p only through $p \bmod 4d$ when $d = \text{sqf}(ab)$.

The Artin property of p is determined by the prime factorisation of $p - 1$: the criterion consults $\ell = 2$ and every prime divisor ℓ of $p - 1$. This is why both decompositions below are

stratified by the *fine signature*

$$\text{sig}(p) = \left(\min(v_2(p-1), 4), \{q \leq 13 : q \mid p-1\}, \min(\omega_{>13}(p-1), 3) \right),$$

a partition of the primes into at most $4 \cdot 32 \cdot 4 = 512$ cells (for odd p one has $v_2(p-1) \geq 1$, so the truncated valuation takes four values, not five). The signature is a coarse summary of the part of $p-1$ that the criterion consults first for small bases; it records neither all prime divisors of $p-1$ nor the power-residue outcomes the criterion tests, and so does not determine Artin status.

4. THE CONSECUTIVE-PRIME EXCLUSION LAW

Definition 1. For an odd prime $p \nmid 10$, write $\left(\frac{10}{p}\right)$ for the Legendre symbol. We say p is an *Artin prime* (for base 10) if the multiplicative order of 10 modulo p equals $p-1$.

If 10 is a quadratic residue mod p then its order divides $(p-1)/2$, so a QR is never a primitive root; $\left(\frac{10}{p}\right) = -1$ is a necessary (not sufficient) condition for the Artin property.

Theorem 2 (Exclusion law for gaps $\equiv 20 \pmod{40}$; cf. [13, Thm. 1.1]). *Let p and $p' = p + g$ be primes with $p, p' > 5$ and $g > 0$, $g \equiv 20 \pmod{40}$. Then*

$$\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right).$$

In particular, 10 is a quadratic residue modulo exactly one of p, p' , and at most one of them is an Artin prime for base 10.

Proof. By multiplicativity, $\left(\frac{10}{p'}\right) = \left(\frac{2}{p'}\right) \left(\frac{5}{p'}\right)$.

Since $g \equiv 0 \pmod{5}$, we have $p' \equiv p \pmod{5}$. By quadratic reciprocity ($5 \equiv 1 \pmod{4}$), $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$ depends only on $p \pmod{5}$; hence $\left(\frac{5}{p'}\right) = \left(\frac{5}{p}\right)$.

Since $g \equiv 4 \pmod{8}$, we have $p' \equiv p + 4 \pmod{8}$. The map $r \mapsto r + 4$ on odd residues mod 8 exchanges $1 \leftrightarrow 5$ and $3 \leftrightarrow 7$. By the second supplementary law, $\left(\frac{2}{p}\right) = +1$ iff $p \equiv \pm 1 \pmod{8}$, and the exchange maps $\{1, 7\}$ onto $\{5, 3\}$ and vice versa; hence $\left(\frac{2}{p'}\right) = -\left(\frac{2}{p}\right)$ for every admissible residue.

Multiplying, $\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right)$. Whichever of the two primes has $\left(\frac{10}{\cdot}\right) = +1$ satisfies $\text{ord}(10) \mid (p-1)/2 < p-1$, so it is not an Artin prime. $\square$

Corollary 3 (Coupling law for gaps $\equiv 0 \pmod{40}$). *If $p, p' = p + g$ are primes with $p, p' > 5$ and $g \equiv 0 \pmod{40}$, then $\left(\frac{10}{p'}\right) = \left(\frac{10}{p}\right)$. In particular, if either is an Artin prime, then 10 is a quadratic non-residue for both.*

Proof. Now $g \equiv 0 \pmod{8}$ and $g \equiv 0 \pmod{5}$, so both $\left(\frac{2}{\cdot}\right)$ and $\left(\frac{5}{\cdot}\right)$ are preserved. $\square$

Remark 4. The modulus 40 is specific to base 10: it is the conductor of the quadratic character $p \mapsto \left(\frac{10}{p}\right)$, namely $4 \cdot 10 = 40$ (the discriminant of $\mathbb{Q}(\sqrt{10})$). For a general non-square integer base a the same argument applies with 40 replaced by the conductor $|d_a|$ of the quadratic character attached to $\mathbb{Q}(\sqrt{a})$: for odd primes p, p' with $p \nmid a$ and $p' \nmid a$, gaps $g \equiv 0 \pmod{|d_a|}$ preserve quadratic-residue status. The coprimality hypothesis is needed, not cosmetic: the primitive character of the field agrees with $\left(\frac{a}{\cdot}\right)$ only away from the primes dividing a , and a prime dividing a square factor of a gives $\left(\frac{a}{p}\right) = 0$ while the field character is nonzero. For

instance $a = 98 = 2 \cdot 7^2$ has conductor 8, and the primes 7 and 23 differ by $16 \equiv 0 \pmod{8}$, yet $\left(\frac{98}{7}\right) = 0$ while $\left(\frac{98}{23}\right) = \left(\frac{2}{23}\right) = +1$. (The base-10 theorem is unaffected: its hypotheses exclude $p = 2, 5$.) Some bases also admit residue classes of gaps that reverse the Legendre symbol on every admissible pair; their existence requires a separate classification. Base 10 is distinguished only by our decimal habit and the full-reptend interpretation.

Remark 5. For gaps in classes other than $0, 20 \pmod{40}$ the relation between $\left(\frac{10}{p}\right)$ and $\left(\frac{10}{p'}\right)$ depends on p as well as g (the shift on $p \pmod{8}$ or $p \pmod{5}$ is not a symbol-preserving or symbol-flipping involution), producing the partial correlations catalogued in Section 8.

We verified Theorem 2 computationally as a guard against error: for all 23,956 consecutive prime pairs below 10^7 with gap $\equiv 20 \pmod{40}$, the Legendre symbols flip in every single case, and a direct order computation on a systematic subsample found no doubly-Artin pair. In the full dataset up to 10^9 (Section 8), the 2,195,882 consecutive pairs with $g \in \{20, 60\}$ contain *no* pair in which both primes are Artin — an empirical count of zero, exactly as the theorem requires.

5. THE SAME-PRIME EXCLUSION LAWS

Throughout, for a positive integer a we write $d_a = \text{sqf}(a)$ for its squarefree part and $\chi_a = \left(\frac{d_a}{\cdot}\right)$ for the quadratic character attached to $\mathbb{Q}(\sqrt{a})$; thus $\chi_a(p) = +1$ exactly when a is a quadratic residue modulo p (for $p \nmid 2a$). We allow a to be a square, in which case $d_a = 1$ and $\chi_a \equiv +1$ is the principal character; this convention costs nothing and lets the statements below dispense with a non-squaredness hypothesis. We have

$$p \in \mathcal{A}_a \implies \chi_a(p) = -1, \quad (1)$$

since a quadratic residue generates a subgroup of index at least 2 in $(\mathbb{Z}/p\mathbb{Z})^\times$.

Everything in this section follows from one line: the quadratic character is completely multiplicative, so for any positive integers a, b and any odd prime $p \nmid ab$,

$$\chi_a(p) \chi_b(p) = \left(\frac{d_a d_b}{p}\right) = \left(\frac{\text{sqf}(ab)}{p}\right) = \chi_{\text{sqf}(ab)}(p), \quad (2)$$

since $d_a d_b$ and $\text{sqf}(ab)$ differ by a square. Combining (2) with the obstruction (1) gives the following.

Theorem 6 (Pair exclusion). *Let a, b be positive integers and set $d = \text{sqf}(ab)$. Then for every odd prime p with $\left(\frac{d}{p}\right) = -1$, the prime p is not simultaneously Artin base a and Artin base b .*

Proof. If $p \mid ab$ then one of the two bases is $\equiv 0 \pmod{p}$ and so is not a primitive root; assume therefore $p \nmid ab$. By (2), $\chi_a(p) \chi_b(p) = \left(\frac{d}{p}\right) = -1$, so $\chi_a(p)$ and $\chi_b(p)$ have opposite signs and in particular one of them equals $+1$; by (1) the corresponding base is not a primitive root. $\square$

The condition $\left(\frac{d}{p}\right) = -1$ is a condition on residue classes. For $d > 1$ the character $\left(\frac{d}{\cdot}\right)$ is non-principal, of fundamental discriminant d if $d \equiv 1 \pmod{4}$ and $4d$ otherwise; in both cases the discriminant divides $4d$, so the excluded set is an explicit union of reduced classes modulo $4d$, and exactly half of those classes carry character -1 . By the prime number theorem in arithmetic progressions the excluded set then has density exactly $1/2$ among the odd primes. So for any two bases with $d > 1$ — no third base required, no special configuration — half of all primes are deterministically barred from being Artin for both, and Theorem 6 is a *congruence* exclusion law.

The excluded case $d = 1$ is exactly $\text{sqf}(a) = \text{sqf}(b)$, i.e. ab a perfect square, i.e. $\chi_a = \chi_b$: then $\chi_a \chi_b = \chi_a^2 \equiv +1$, nothing is ever barred, and Theorem 6 is vacuous rather than false. (For instance $a = 2$, $b = 8$: the two bases generate the same quadratic field, and indeed 2 and 8 are simultaneously primitive roots for $p = 3, 5, 11, 29, 53, \dots$) Such pairs are precisely the ones our base set avoids: all twelve bases are squarefree and distinct, so $d > 1$ for all 66 pairs. Writing the classes out in the four cases we return to below:

$$\begin{aligned} (2, 6), d = 3: p \equiv 5, 7 \pmod{12}; & \quad (3, 6) \text{ and } (5, 10), d = 2: p \equiv 3, 5 \pmod{8}; \\ (2, 10), d = 5: p \equiv 2, 3 \pmod{5}. \end{aligned}$$

We verified each over all primes below 6×10^4 with no exceptions, and Section 11 verifies the theorem itself against all 50,847,524 primes below 10^9 for all 66 pairs of our base set simultaneously. Theorems 6 and 7 are also machine-checked in Lean 4 against Mathlib, including the step (1) and the (5, 10) instance above, whose barring condition is supplied as $p \bmod 8$ (the public statement of that instance retains its nonvanishing hypothesis rather than deriving it), and each checked statement depends only on a subset of `propext`, `Classical.choice` and `Quot.sound`. Every checked statement depends on no axiom beyond `propext`, `Classical.choice` and `Quot.sound`. We state the limit of this plainly: the general translation of $\left(\frac{d}{p}\right) = -1$ into residue classes modulo $4d$ is *not* formalised generically, but carried out by hand per instance. See `LEAN.NOTE.md` in the accompanying package.

The triple form of the law, which was our original route to it, is the special case in which the barring character is supplied by a third base of the set.

Theorem 7 (Triple exclusion). *Let a, b, c be positive integers with $\text{sqf}(c) = \text{sqf}(ab)$ and let p be an odd prime. Then p is not simultaneously Artin base a , Artin base b , and Artin base c .*

Proof. If p is Artin base c then $\chi_c(p) = -1$ by (1), and $\chi_c = \chi_d$ with $d = \text{sqf}(ab) = \text{sqf}(c)$; now apply Theorem 6. $\square$

Corollary 8. *Within any multiplicative triple — for instance $(2, 5, 10)$, $(3, 5, 15)$, $(3, 7, 21)$, $(2, 3, 6)$ — an odd prime can be Artin for at most two of the three bases. The restriction to odd p is necessary and not cosmetic: every odd integer is vacuously a primitive root modulo 2, so the prime 2 is Artin base 3, base 5 and base 15.*

The gain in passing from Theorem 7 to Theorem 6 is not cosmetic, though it is a gain in usable content rather than in logical strength: both are pointwise statements at every odd prime. The difference is in what one must know to apply them. To invoke Theorem 7 one must first identify a third base c with $\text{sqf}(c) = \text{sqf}(ab)$ and establish that c is a primitive root at the prime in question — itself an instance of the Artin problem, conjectural in general and (under GRH) of positive but base-dependent density. Theorem 6 needs neither: the barring condition is a congruence, decidable from p alone by reciprocity, and for $d > 1$ it holds on a set of density exactly $1/2$. The triple law is the shadow that the pair law casts when one insists on reading the barring character off a primitive root instead of off a residue class.

Remark 9 (Sharpness of the hypotheses). In both theorems no hypothesis on the bases beyond positivity is needed, only “ p odd” (together with the barring condition $(d/p) = -1$ in Theorem 6). Non-squareness of the bases is not required: if a is a square then $\chi_a(p) = +1$ and a is not a primitive root, so the conclusion is immediate. The same remark disposes of $c = 1$ in Theorem 7, which arises whenever $\text{sqf}(a) = \text{sqf}(b)$. The hypothesis $p \neq 2$ must however be kept, and is not removable: $(\mathbb{Z}/2\mathbb{Z})^\times$ is trivial, so every odd integer is vacuously a primitive root mod 2 and e.g. $(a, b, c) = (3, 5, 15)$ violates the conclusion there. The accompanying Lean formalisation (Section 5) proves the character-parity core of both theorems together with selected

concrete instances; the statements it checks retain explicit nonvanishing hypotheses on the bases (discharged in those instances), and it does not formalise a generic squarefree-part-to-witness bridge.

Remark 10. The exclusion is sharp: primes Artin for exactly two of $(2, 5, 10)$ exist in abundance. The smallest is $p = 3$, where 2 and 5 are both primitive roots (each of order $2 = p - 1$) while $10 \equiv 1$ is not; the smallest for which 10 itself is Artin is $p = 7$, where 5 and 10 are both primitive roots (each of order 6) while 2 has order 3; likewise $p = 23$, where 5 and 10 are primitive roots and 2 is a quadratic residue of order 11. More systematically, in our data the pair $(2, 10)$ has $P(\mathcal{A}_{10} \mid \mathcal{A}_2) = 0.355$: two-of-three is common, three-of-three is impossible.

Remark 11. Theorem 6 is the same-prime analogue of the gap exclusion law for consecutive primes proved above (Theorem 2); the arbitrary-base form is [18]: both are instances of a parity obstruction in the group generated by the quadratic characters involved, and in both the excluded set is a union of residue classes. There the two characters were χ_a evaluated at p and at $p + g$, and the obstruction was arranged by the shift g ; here the characters belong to different bases at the same prime, and the obstruction is arranged by multiplicative dependence. The Galois-theoretic density computations of Matthews [24] and Moree–Stevenhagen [23] contain this relation as the vanishing criterion recalled in Section 1 (the degree of the compositum of the relevant Kummer extensions drops); the elementary pointwise form seems worth recording.

6. DATA AND COMPUTATION

The two axes of this paper use two censuses over the same primes, with slightly different exclusions at the low end, by design. We describe each.

The consecutive-prime census (base 10). For every prime $7 \leq p \leq 10^9$ we compute: the gap to the neighbouring primes, the number $\omega(p - 1)$ of distinct prime factors of $p - 1$ (by trial division), the Artin indicator (computed exactly via $10^{(p-1)/q} \not\equiv 1 \pmod{p}$ for each prime $q \mid p - 1$, using GMP modular exponentiation), and small residues of p . A segmented sieve of Eratosthenes generates the primes; the corrected sieve emits all 50,847,531 primes from 7 to 999,999,937 (the largest prime $\leq 10^9$).

Independent recomputation. The census has been recomputed from scratch by a second, deliberately disjoint implementation, which shares no code with the sieve above: it uses hand-written 128-bit modular exponentiation in place of GMP, and it factors $p - 1$ by a segmented sieve of factors over the shifted interval rather than by per-prime trial division, so that an error in either implementation is unlikely to be mirrored in the other. Both were first checked against **sympy** at 10^6 , where all three routes agree exactly. At 10^9 the independent implementation reproduces the published census in every quantity: 50,847,531 primes, 19,016,617 Artin primes, the joint table $\begin{bmatrix} 19758045 & 12072868 \\ 12072869 & 6943748 \end{bmatrix}$, $\delta = -0.014140158840$, and zero doubly-Artin pairs among all 2,195,882 pairs at gaps 20 and 60.

The same program extends the census to 10^{10} (455,052,508 primes) and 10^{11} (4,118,054,810 primes). In both cases the prime count equals $\pi(x)$ less the three excluded primes 2, 3, 5, and the Artin density agrees with Artin’s constant 0.3739558 to five decimal places (0.373955 and 0.3739543). Theorem 2 continues to hold: there are no doubly-Artin pairs among the 20,700,958 pairs at gaps 20, 60 below 10^{10} , nor among the 194,296,748 such pairs below 10^{11} . The computation is run as a few hundred contiguous shards and stitched, with the pairs straddling shard boundaries added back; re-sharding 10^{10} under a different partition reproduces every quantity identically, including δ to fifteen decimal places, so the totals do not depend on the shard layout.

Endpoint correction. An earlier version of the sieve contained two endpoint bugs: it emitted a duplicate row for $p = 7$ and omitted the final prime $p = 999,999,937$. These have been corrected. The duplicate was already skipped by the analysis scripts; the missing endpoint adds one pair at gap 8 (both primes non-Artin: $\text{ord}_{10}(999,999,937) = 333,333,312$; independently verified via `sympy`). This correction changes the pair count from 50,847,529 to 50,847,530 and does not materially alter any displayed summary statistic.

As a global check, the number of Artin primes in the dataset is 19,016,617, a proportion 0.373993, matching Artin's constant 0.373956 to four decimal places (for base 10 the squarefree part is $10 \not\equiv 1 \pmod{4}$, so no correction factor applies and the conjectural density is exactly C ; see [11, 15]). Agreement with a conjectural density is a sanity check, not a verification of every indicator.

The consecutive-pair analysis is a single streaming pass over the 50,847,530 ordered pairs (p_n, p_{n+1}) , accumulating 2×2 contingency tables of $(\mathbf{1}[\text{Art}_n], \mathbf{1}[\text{Art}_{n+1}])$: globally, stratified by gap g , and stratified by joint residues of (p_n, p_{n+1}) to various moduli. All counts reported below are exact (no sampling), except where noted. Analysis scripts and their outputs are available with the data; see the Data Availability statement.

Runtime, hardware, and validation. The computations to 10^9 were performed on a single machine with a 2-core Intel Xeon E5-2673 v4 processor at 2.3 GHz and 8 GB of RAM, running Ubuntu Linux, using no parallelism beyond a single core. The independent recomputation and the extensions to 10^{10} and 10^{11} were run on a separate 32-core AMD machine (gcc 15.2.0) as a few hundred contiguous shards; that run is described above and uses a different implementation. The segmented sieve (segment length 2^{21} , so the working set stays within cache) together with the per-prime computation — trial-division factorisation of $p - 1$ and the Artin test by modular exponentiation — generates the full dataset to 10^9 ; peak memory usage is under 200 MB, as only one segment and the small primes to $\sqrt{x}$ are held in memory. The streaming pair analysis reads the dataset once and runs in minutes. As validation we checked: (i) the prime count $\pi(10^9) = 50,847,534$ against the known value; (ii) the Artin prime count 19,016,617 (proportion 0.373993) against Artin's constant 0.373956, as discussed above; (iii) Theorem 2 on all 23,956 pairs with gap $\equiv 20 \pmod{40}$ below 10^7 , with an independent `sympy` order computation on a subsample; and (iv) the mod-12 consecutive-residue transition matrix against the values reported in [14] at comparable scale.

The cross-base census. By a segmented sieve we enumerated all primes $p \leq 10^9$ (50,847,534 primes). The programs discard only the smallest primes, for which the 128-bit Euler-criterion arithmetic and the signature encoding are degenerate: $p < 5$ for the correlation and decomposition runs, leaving 50,847,532 primes, and $p < 31$ for the signature-and-character run of Section 13, leaving 50,847,524. A prime dividing a base is *not* discarded; it is simply recorded as non-Artin for that base, which is correct, since 0 is not a primitive root. The two counts therefore differ by exactly the eight primes in $[5, 31]$. For each prime we factored $p - 1$ once by trial division and computed the Artin indicator for each of the twelve bases by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p - 1$, in 128-bit arithmetic). One streaming pass accumulates, for each of the 66 unordered base pairs, the 2×2 contingency table of joint Artin status — globally, and stratified by (i) $\omega(p - 1)$ and (ii) the fine signature

$$\text{sig}(p) = \left(\min(v_2(p - 1), 4), \{q \leq 13 : q \mid p - 1\}, \min(\omega_{>13}(p - 1), 3) \right),$$

a partition of the primes into at most $4 \cdot 32 \cdot 4 = 512$ cells (for odd p we always have $v_2(p - 1) \geq 1$, so the truncated valuation takes four values, not five). The signature is a coarse summary of the part of $p - 1$ that the Artin criterion consults first for small bases — the 2-part, the small

odd prime divisors, and a truncated count of the larger factors. It records neither all prime divisors of $p - 1$ nor the power-residue outcomes that the criterion actually tests, and so does not determine Artin status: the primes 139 and 223 have the same signature and the same $(\frac{10}{p}) = -1$, yet $\text{ord}_{139}(10) = 46$ and $\text{ord}_{223}(10) = 222$. A separate run, also to 10^9 (50,847,524 primes), stratified each pair additionally by the quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ of the two bases on top of the signature. The whole computation runs in under two hours on a single core of a commodity machine (2-core Intel Xeon E5-2673 v4, 8 GB RAM); all counts are exact.

As validation, the marginal Artin densities agreed with the Hooley densities to within 6×10^{-5} absolute (0.014% relative) for every base — e.g. 0.373957 for base 2, against Artin’s constant 0.373956, and 0.393642 for base 5, whose density carries the classical correction factor $C \cdot \frac{20}{19}$ for $d \equiv 1 \pmod{4}$ (see [11, 15]).

7. THE GLOBAL ANTICORRELATION

TABLE 1. Artin status of consecutive primes, all 50,847,530 pairs with $7 \leq p_n < p_{n+1} \leq 10^9$.

Quantity	Value
$P(\text{Art}_{n+1})$	0.373993
$P(\text{Art}_{n+1} \mid \text{Art}_n)$	0.365141
$P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$	0.379281
$\delta = \text{difference}$	−0.014140
ϕ coefficient	−0.014140
χ^2 (1 df)	10,167
signed $\sqrt{\chi^2}$ (descriptive)	−100.8

Table 1 presents the central fact: an Artin prime is followed by another Artin prime about 1.4 percentage points less often than a non-Artin prime is. The Pearson statistic is 10,167 (signed square root −100.8 under the nominal independent-status reference model). This is a deterministic finite census, not a random survey; the statistic is reported as a descriptive measure of the strength of association, and should not be interpreted as a sampling confidence level without specifying and justifying a null stochastic model. Consecutive pairs overlap, and residue-cell memberships are serially structured.

The sign is what the Lemke Oliver–Soundararajan repulsion predicts. Heuristically: the Artin property requires $(\frac{10}{p}) = -1$, which is a condition on $p \pmod{40}$; it is further favoured by $3 \nmid p-1$, i.e. $p \equiv 2 \pmod{3}$, since a factor of 3 in $p-1$ contributes the largest single suppression $(1 - \frac{1}{3})$ of the primitive-root density among odd primes. Consecutive primes repel in their residues modulo 40 and modulo 3 [14]; conditions favourable to the Artin property therefore tend *not* to repeat at p_{n+1} when they hold at p_n , depressing $P(\text{Art}_{n+1} \mid \text{Art}_n)$.

For reference, the mod-12 transition matrix in our data displays the repulsion prominently: the diagonal entries $P(p_{n+1} \equiv r \mid p_n \equiv r)$ are 0.1813–0.1815 for $r \in \{1, 5, 7, 11\}$, against the equidistribution value 0.25 — a 27% deficit, consistent in magnitude with the computations in [14] at comparable scales.

8. GAP STRUCTURE

Stratifying by the gap $g = p_{n+1} - p_n$ reveals sign-alternating within-gap effects that are individually far larger in magnitude than the global $\delta = -0.0141$. The global value is *not*

the pair-frequency-weighted average of the within-gap values: for a marginal conditional risk difference such as δ , the strata are additionally reweighted by the gap-conditional distribution of Art_n , so no such averaging identity holds. Indeed the pair-weighted mean of the within-gap differences over the 36 qualifying gap classes (50,114,759 pairs, 98.6% of the census) is +0.00159, opposite in sign to the global value. The two quantities answer different questions, and the gap table below should be read as a decomposition of *where* the association lives, not as a summand decomposition of δ itself (cf. the caution in Section 9). Table 2 shows all gaps with at least 10^5 pairs up to $g = 60$; additional qualifying gaps ($g \in \{62, 64, 66, 70, 72, 78\}$) are included in the analysis scripts and archived results but omitted from the table for brevity. Figure 1 displays $\delta(g)$ with the residue class of $g \bmod 40$ marked, including all qualifying gaps.

TABLE 2. Conditional Artin dependence by gap ($N \geq 10^5$ pairs, $g \leq 60$).
 $\delta(g) = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$.

g	N	$P(A A)$	$P(A \neg A)$	$\delta(g)$	g	N	$P(A A)$	$P(A \neg A)$	$\delta(g)$
2	3,424,504	0.291	0.293	-0.002	32	683,896	0.176	0.386	-0.209
4	3,424,679	0.562	0.381	+0.181	34	719,531	0.445	0.445	+0.001
6	6,089,791	0.407	0.382	+0.025	36	1,171,524	0.559	0.314	+0.245
8	2,695,109	0.160	0.409	-0.249	38	548,746	0.288	0.293	-0.006
10	3,484,767	0.448	0.443	+0.005	40	648,780	0.899	0.257	+0.643
12	4,468,957	0.512	0.283	+0.229	42	954,456	0.380	0.358	+0.023
14	2,464,565	0.298	0.296	+0.002	44	389,634	0.344	0.249	+0.095
16	1,846,097	0.296	0.543	-0.247	46	334,720	0.469	0.469	+0.000
18	3,351,032	0.378	0.359	+0.019	48	577,247	0.216	0.454	-0.238
20	1,824,043	0.000	0.553	-0.553	50	328,066	0.311	0.307	+0.005
22	1,569,679	0.444	0.447	-0.004	52	245,804	0.572	0.381	+0.191
24	2,367,474	0.295	0.411	-0.116	54	410,754	0.382	0.356	+0.026
26	1,119,585	0.313	0.313	+0.000	56	211,462	0.225	0.389	-0.164
28	1,219,243	0.619	0.366	+0.253	58	181,948	0.437	0.441	-0.004
30	2,177,991	0.390	0.363	+0.027	60	371,839	0.000	0.592	-0.592

Three features stand out.

(i) *The exclusion rows.* At $g = 20$ (1,824,043 pairs) and $g = 60$ (371,839 pairs), $P(A \mid A)$ is exactly zero: among 2.2 million consecutive pairs, not one is doubly Artin. This is Theorem 2 in action, and conversely the observation that motivated its proof.

(ii) *The $g = 40$ spike.* $P(A \mid A) = 0.899$, the largest conditional probability in the table and 2.4 times the unconditional density. Two forced mechanisms combine. First, by Corollary 3, if p_n is Artin then 10 is a non-residue for p_{n+1} as well, a necessary condition for the Artin property. Second, $40 \equiv 1 \pmod{3}$ forces the pair's mod-3 residues: $p_n \equiv 2 \pmod{3}$ would give $3 \mid p_n + 40$, impossible; so every $g = 40$ pair has $p_n \equiv 1, p_{n+1} \equiv 2 \pmod{3}$ — that is, $3 \mid p_n - 1$ (suppressing Art_n) and $3 \nmid p_{n+1} - 1$ (favouring Art_{n+1}). Conditioning on Art_n therefore selects pairs in which p_{n+1} enjoys both the non-residue status and the mod-3 advantage. These conditions are consistent with the large observed value 0.899 but do not quantitatively derive it without an explicit density calculation.

(iii) *Sign classes.* Several gaps that are divisible by 3 but not constrained mod 40, for instance 6, 18, 30, 42 and 54, show a small positive $\delta \approx +0.02$. A natural guess is that the pair shares its mod-3 status, weakly coupling the largest single suppression factor. This is a selected family rather than a rule: other gaps divisible by 3 and lying in neither 0 nor 20 mod 40, such as 12, 24 and 36, do not all behave this way, and we have no statement covering every class.

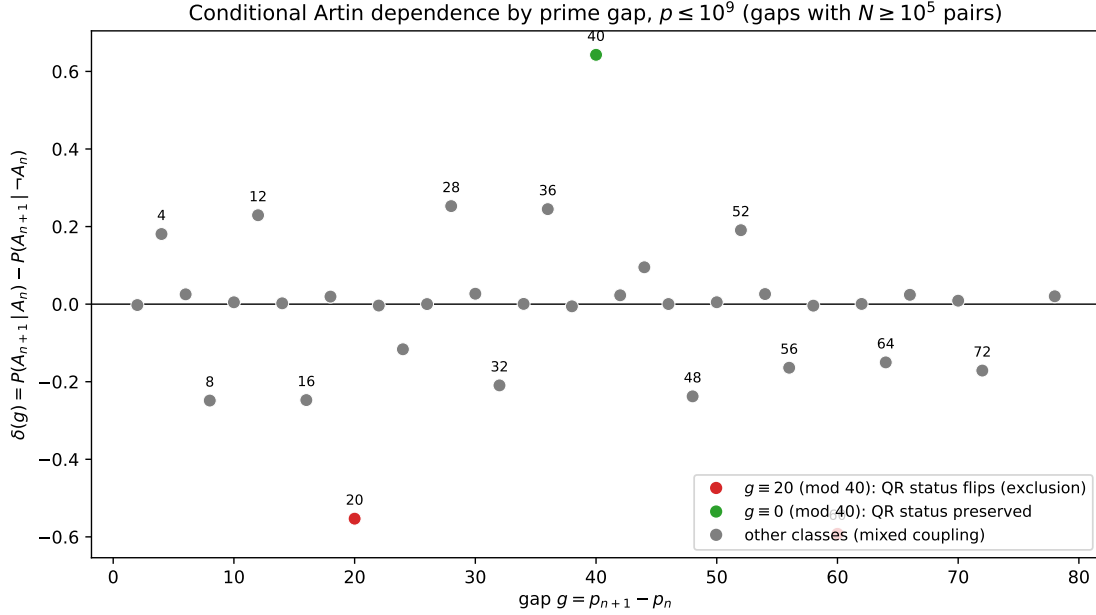


FIGURE 1. $\delta(g)$ for all gaps with $N \geq 10^5$ pairs, $p \leq 10^9$. Red: $g \equiv 20 \pmod{40}$ (Theorem 2 forces $P(A|A) = 0$). Green: $g \equiv 0 \pmod{40}$ (Corollary 3 preserves quadratic-residue status). Grey: other classes.

Some gaps with adverse mod-8 and mod-5 shifts show strong negative δ . We can therefore offer only a qualitative reading of Table 2 in terms of how the shift by g permutes the residues of p modulo 8, 5 and 3; we do not derive the entries, and outside the exclusion classes nothing here is forced.

9. RESIDUE CONDITIONING

To what extent does the association persist once the explicit residue channels are held fixed? We treat what follows as an exploratory conditioning diagnostic; it does not answer that question quantitatively. We condition on the *joint* residue class of (p_n, p_{n+1}) to modulus m and measure the residual within-cell dependence. The statistic reported is the sum of per-cell Pearson χ^2 values over all non-degenerate cells meeting a minimum count threshold, together with the pair-weighted mean of per-cell δ . This is a descriptive aggregate of per-cell associations; it is *not* the standard Cochran–Mantel–Haenszel statistic (which would be a single one-degree-of-freedom test of a common association).

Conditioning on $m = 120 = \text{lcm}(8, 3, 5)$ fixes the quadratic character $\left(\frac{10}{\cdot}\right)$ of both primes (its conductor 40 divides 120) and the mod-3 channel. Cells in which 10 is a quadratic residue for p_n or p_{n+1} are degenerate for the analysis (one margin of the 2×2 table vanishes — the Artin property is impossible); the informative population is roughly a quarter of all pairs, those in doubly-non-residue cells. The selected populations in Table 3 are the thresholded subsets of this population: 12,255,203 pairs (24.10% of the census) at modulus 120 and 12,086,765 pairs (23.77%) at modulus 840. The unthresholded doubly-non-residue fraction need not be exactly $1/4$.

Interpretation and limitations. The three statistics in Table 3 have successively smaller absolute values: -0.0141 (global) then -0.00106 (mod 120, 145 cells covering 12.3 million

TABLE 3. Selected-cell residual association after conditioning on joint residues mod m . “ δ_{res} ” is the pair-weighted mean within-cell δ over non-degenerate cells meeting the count threshold; “cells” is the number of such cells; “pairs used” is the total pairs in those cells. Cells with fewer than 5000 ($m = 120$) or 2000 ($m = 840$) pairs are excluded.

Conditioning	cells	pairs used	$\sum \chi^2$ (df)	δ_{res}
none (global)	1	50,847,530	10,167 (1)	−0.014140
mod 120 joint residues	145	12,255,203	789.2 (145)	−0.001055
mod 840 joint residues	633	12,086,765	719.8 (633)	−0.000473

pairs) then −0.00047 (mod 840, 633 cells covering 12.1 million pairs). We emphasise that these are three different quantities computed on three different populations, not one quantity being reduced; the comparison is an exploratory conditioning diagnostic, not a decomposition. Note also that a signed mean can be small because sizable positive and negative within-cell associations cancel, so smallness in this metric is not by itself evidence that dependence has gone. The summed per-cell χ^2 at mod 840 is 719.8 on 633 nominal degrees of freedom ($\chi^2/\text{df} = 1.14$); we do not read $\chi^2/\text{df} \approx 1$ as the disappearance of dependence, since no reference distribution has been established for this serially structured, selected collection. Two further observations: (i) the three values decrease in magnitude in the order stated (−0.0141, −0.00106, −0.00047); (ii) a split-sample check (pairs with $p < 5 \times 10^8$ versus $p \geq 5 \times 10^8$, conditioning mod 120) gives residuals −0.00096 and −0.00094 — stable across the range.

Several caveats apply. The global δ over 50.8 million pairs and the selected-cell δ_{res} over 12.1 million selected informative pairs are *different estimands on different populations*; arithmetic comparison of their magnitudes is descriptive but does not constitute a proper decomposition. A marginal risk difference is not, in general, the pair-weighted mean of stratum risk differences without a between-group term and specified weights. The per-cell Pearson calibration assumes independent observations within each cell, which is not established for this serially structured deterministic census. The paper does not specify a numerical model for the omitted channels against which “compatible” could be formally tested. Those omitted channels are not only the divisibility channels at $q \geq 11$: fixing p modulo 840 determines whether 3, 5 and 7 divide $p - 1$, but not whether $10^{(p-1)/q} \equiv 1 \pmod{p}$ when one of them does, so cubic and higher power-residue obstructions at these same small primes are also unmodelled. The mod-3 channel here is the divisibility channel, not the full cubic-residue condition.

Joint-residue conditioning yields a smaller selected-cell signed residual association than the global figure. The remaining association is small in that metric but is not established to vanish. These results are consistent with a substantial role for arithmetic residue structure; they do not constitute a complete decomposition or an independent quantitative verification of the Hardy–Littlewood-based explanation of the consecutive-prime bias.

10. THE INTERMEDIATE LINK: REPULSION OF $\omega(p - 1)$

The chain from residue bias to Artin status passes through the factorisation of $p - 1$. Its first summary statistic is $\omega(p - 1)$, the number of distinct prime factors, which has normal order $\log \log p$ over shifted primes $p - 1$ — a classical result of Erdős [4]. (We invoke the normal order only; we do not use a mean-value asymptotic, which would need separate tail control.) If consecutive primes repel modulo every small q , then $q \mid p_n - 1$ makes $q \mid p_{n+1} - 1$ *less* likely

than average, and summing over q one predicts a negative correlation between $\omega(p_n - 1)$ and $\omega(p_{n+1} - 1)$.

The data confirm this with a value that is, to our knowledge, unreported:

$$r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.0410 \quad (50,847,530 \text{ pairs}, p \leq 10^9).$$

The factorisations of $p - 1$ for consecutive primes are therefore not independent. We stress that this measurement establishes dependence, not its cause. The heuristic above accounts only for the same- q terms: writing $I_q(p) = \mathbf{1}_{q|p-1}$,

$$\text{Cov}(\omega(p_n - 1), \omega(p_{n+1} - 1)) = \sum_q \text{Cov}(I_q(p_n), I_q(p_{n+1})) + \sum_{q \neq r} \text{Cov}(I_q(p_n), I_r(p_{n+1})),$$

and the cross terms $q \neq r$ are neither estimated nor signed here. Summing fixed-modulus predictions over all q up to a growing cutoff would in any case require uniformity and tail control that we do not supply. Nor is ω a sufficient statistic for Artin status: which primes divide $p - 1$, and whether 10 is a q th power residue, both matter. The observed repulsion is consistent with the same residue-class repulsion that governs the primes, but we do not establish that it has the same cause. The Pearson coefficient is reported as a descriptive nominal value.

11. THE CROSS-BASE CORRELATION

For each pair write $\phi(a, b)$ for the Pearson correlation of the indicator variables (the ϕ coefficient of the 2×2 table). This is an exhaustive census of a fixed interval for a deliberately chosen finite set of bases, so the coefficients below are exact and carry no sampling error. We therefore report effect sizes and their variation with x , and avoid the vocabulary of statistical inference. Where a scale factor is quoted we write $\sigma_0 = N^{-1/2} \approx 0.00014$ at $N \approx 5.08 \times 10^7$: this is the standard deviation of ϕ under an i.i.d. independence benchmark that does *not* describe these data — the primes are enumerated, not sampled — and ratios ϕ/σ_0 should be read as model-relative diagnostics indicating that the effects are far larger than that benchmark's scale, not as tests of any hypothesis. No claim below depends on such a ratio.

One convention should be stated, since it affects the conditional averages in Sections 12 and 13. Within-cell ϕ is undefined when a cell is constant in one of the two indicators (for example cells with $5 \mid p - 1$, in which no prime is Artin base 5). Such cells carry zero conditional covariance and are omitted from the pooled average, which therefore draws on 73% of primes on average for the signature stratification, and 23% for the finer signature-plus-character stratification, where three of the four character quadrants are identically non-Artin. Explicitly, for a base pair (a, b) we report

$$\phi \mid \text{sig} = \frac{\sum_{s \in E_{ab}} n_s \phi_{ab,s}}{\sum_{s \in E_{ab}} n_s},$$

where s runs over signature cells, n_s is the number of primes in cell s , and E_{ab} is the set of cells with $n_s \geq 200$ in which *both* indicators vary. This is a prime-count-weighted descriptive mean of within-cell correlations, not a partial correlation coefficient and not an additive component of ϕ ; outer means over the 66 base pairs are unweighted. Pooling instead on the covariance scale over all cells, which requires no such exclusion, gives 95.3% removal, in agreement with Table 6.

Three observations, in increasing order of depth.

Observation 12 (Positivity). $\phi(a, b) > 0$ for 64 of the 66 pairs. Artin statuses of different bases at the same prime *attract*.

TABLE 4. Same-prime cross-base Artin correlation at $x = 10^9$: summary and extremes. “Triple-completed” means the base set contains a third base c with $\text{sqf}(c) = \text{sqf}(ab)$. This is a property of the chosen base set, not of the pair: the two bases of such a pair are multiplicatively independent in the usual sense, and adding or removing c changes the label but not the pair’s density, character or Kummer field.

	pairs	mean ϕ
all pairs	66	+0.0352
non-completed	51	+0.0363
triple-completed	15	+0.0314
largest: (5, 13)		+0.0667
smallest: (3, 15)		−0.0303
second smallest: (2, 10)		−0.0302

TABLE 5. The ten largest $|\phi|$ at $x = 10^9$, with the values after conditioning on $\omega(p-1)$ and on the fine signature $\text{sig}(p)$ (pair-weighted mean of within-cell ϕ ; cells with fewer than 200 primes excluded). “TC” marks triple-completed pairs.

a	b	ϕ	$\phi \mid \omega$	$\phi \mid \text{sig}$	TC
5	13	+0.0667	+0.0200	+0.0003	
5	17	+0.0649	+0.0218	+0.0025	
5	29	+0.0631	+0.0246	+0.0007	
5	10	+0.0624	+0.0285	−0.0125	✓
5	6	+0.0623	+0.0292	+0.0005	
5	11	+0.0623	+0.0300	+0.0004	
5	15	+0.0622	+0.0297	−0.0255	✓
3	5	+0.0622	+0.0284	+0.0015	✓
5	7	+0.0622	+0.0297	+0.0002	
2	5	+0.0621	+0.0336	+0.0004	✓

The sign is the opposite of the consecutive-prime correlation measured above (§7, where $\delta < 0$), and the mechanism is transparent: there the two primes had *different* (and LOS-anticorrelated) values of $p-1$; here every base is tested against the *same* $p-1$. If $p-1$ has few small prime factors and a small 2-part, the primitive-root condition is easier for every base at once. Formally, the Artin indicators share the common cause $\text{sig}(p)$.

Observation 13 (The negative pairs are triple-completed). The only pairs with $\phi < 0$ are (2, 10) and (3, 15): in each, the product of the two bases is, up to a square, the third base of a multiplicative triple contained in the set $(2 \cdot 10 = 4 \cdot 5, 3 \cdot 15 = 9 \cdot 5)$. For (2, 10): $P(\mathcal{A}_{10} \mid \mathcal{A}_2) = 0.3551$ versus $P(\mathcal{A}_{10} \mid \neg \mathcal{A}_2) = 0.3853$.

The sign reversal is where Theorem 6 becomes visible statistically. For every pair, triple-completed or not, the primes split into two halves according to the sign of $\chi_d(p)$, $d = \text{sqf}(ab)$: on the half with $\chi_d(p) = -1$ the values $\chi_a(p)$ and $\chi_b(p)$ are forced *opposite*, so joint Artin status is impossible there, while on the half with $\chi_d(p) = +1$ they are forced *equal* and joint Artin status is unobstructed. This is visible directly in the data: over all 66 pairs and all 50,847,524

TABLE 6. Mean cross-base correlation after conditioning, all 66 pairs at $x = 10^9$. “Removed” is $1 - |\phi_{\text{cond}}|/|\bar{\phi}|$, the means taken signed over the 66 pairs; on the mean-absolute scale the corresponding figures are 51% and 87%, and on the covariance scale 57% and 95%.

conditioning	mean ϕ	removed
none	0.0352	—
$\omega(p-1)$	0.0158	55%
$\text{sig}(p)$: $(v_2, \{q \leq 13\}, \omega_{>13})$	0.0019	95%

primes, the number of primes Artin for both bases with $\chi_a(p) \neq \chi_b(p)$ is exactly zero, while the conditional joint density on the complementary half is 0.299.

What then distinguishes the two negatively correlated pairs is not the existence of the exclusion — every one of the 66 pairs has it, on exactly half the primes — but the identity of the barring character χ_d , and in particular how its conductor interacts with the common cause $p-1$. Among the fifteen triple-completed pairs, $d = \text{sqf}(ab)$ takes the values $\{2, 3, 5, 6, 7, 10, 15, 21\}$, and the two pairs with $d = 5$ — $(2, 10)$ and $(3, 15)$, of conductor 5, the smallest conductor occurring and the only odd prime one — are exactly the two with $\phi < 0$. That is the sharp empirical discriminant: not dependence, which thirteen positively correlated pairs also enjoy, but $\text{cond}(\text{sqf}(ab)) = 5$. A plausible reading is that a small conductor makes the barring condition $\chi_d(p) = -1$ resolve on a coarse residue class that is comparatively well correlated with the divisibility structure of $p-1$ driving the common cause, so the exclusion survives the averaging instead of being diluted by it; we do not have a proof of this and record it as an observation. Section 13 tests the reading by conditioning the common cause away, after which the exclusion becomes visible on six triple-completed pairs rather than two.

12. DECOMPOSITION: THE CORRELATION IS THE STRUCTURE OF $p-1$

Conditioning on $\omega(p-1)$ alone — the crude size of the factorisation — removes barely half of the correlation. This is worth emphasising because ω is the statistic through which the consecutive-prime papers routed the mechanism, and it is genuinely insufficient here: two primes with the same $\omega(p-1)$ can present very different Artin conditions (it matters *which* primes divide $p-1$, and with what 2-adic valuation, not how many). The fine signature — 2-adic valuation capped at 4, the exact set of primes ≤ 13 dividing $p-1$, and the count of larger prime factors capped at 3 — is the information the Artin criterion actually consults for small bases, and conditioning on it removes 95% of the mean correlation (Table 6). That the fine structure of $p-1$, rather than its coarse size, is what governs Artin behaviour is in the same spirit as the refined order-distribution conjectures of [19], which pass from the Artin condition $\text{ord}_p(a) = p-1$ to the distribution of $\omega((p-1)/\text{ord}_p(a))$ — finer information about the index than its mere vanishing, just as our signature is finer information about $p-1$ than $\omega(p-1)$.

We regard this as the main statistical finding: *cross-base Artin correlation at a single prime is, to first order, not an interaction between the bases at all*. It is a common-cause effect of the shared integer $p-1$. A model that reproduces the observed within-signature conditional marginals and the observed signature frequencies — equivalently, one in which the two indicators are conditionally independent given the signature — accounts for 95% of the observed mean association. Conditional independence alone is not enough: an independent model with constant cell probabilities is an immediate counterexample.

TABLE 7. Residual correlation after conditioning on $\text{sig}(p)$ and after additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$, split by triple-completion. All values at $x = 10^9$ over 50,847,524 primes.

group	mean ϕ	mean $\phi \mid \text{sig}$	mean $\phi \mid \text{sig, QR}$
triple-completed (15 pairs)	+0.0314	+0.0049	+0.0049
mean <i>absolute</i> residual		0.0179	
non-completed (51 pairs)	+0.0363	+0.0010	+0.0052
mean <i>absolute</i> residual		0.0010	
all 66 pairs	+0.0352	+0.0019	+0.0052

13. THE RESIDUAL AND THE EXCLUSION LAW

The 5% that survives is not uniformly distributed: it is concentrated on the triple-completed pairs. This section argues that it is the statistical trace of Theorem 6 — the deterministic exclusion that the unconditional correlation of Section 11 had largely averaged away.

Two facts pin down the mechanism, both measured over the full 10^9 range.

First, after fine conditioning the triple-completed pairs retain substantial residuals, and every pair with a substantial *negative* residual is triple-completed. The six pairs below are the only ones with $\phi \mid \text{sig} < -0.004$, and all six are triple-completed. We state this as the exact combinatorial fact it is — six of the six most negative residuals fall in the triple-completed class — rather than attaching a p -value to it: the 66 pairs share endpoints and the base set was chosen to contain multiplicative triples, so the pair labels are neither independent nor exchangeable and no randomisation model licences such a test.

$$\begin{aligned}
 (2, 10): -0.0343, & \quad (5, 15): -0.0255, & \quad (6, 10): -0.0144, \\
 (5, 10): -0.0125, & \quad (7, 21): -0.0060, & \quad (10, 15): -0.0049.
 \end{aligned}$$

At $x = 10^9$ these are also the only six negative residuals of any size, but that sharper statement is not stable with x : at $x = 3 \times 10^7$ a further twelve pairs have residuals marginally below zero, all within $3\sigma_0$ of it in the benchmark scale of Section 11.

The residual is not sign-definite, and we record the exceptions explicitly. Nine of the fifteen triple-completed pairs have *positive* residuals: eight of them small, and one, $(2, 6)$, a large outlier at $+0.1598$ — the largest residual of any pair in the collection. Its mechanism is transparent and is the same entanglement phenomenon seen from the other side: $\text{sqf}(6) = 2 \cdot 3$. The relevant family is narrower than $3 \nmid p-1$ alone, however. If $3 \nmid p-1$ and $v_2(p-1) \geq 2$ then $p \equiv 5 \pmod{12}$, so $(3/p) = -1$ and Theorem 6 *forbids* 2 and 6 together. Agreement can occur only on the complementary family $v_2(p-1) = 1$, that is $p \equiv 11 \pmod{12}$, where $(3/p) = +1$ and the two quadratic characters coincide; that the remaining odd-prime tests then also very nearly coincide is what we measure, not something the factorisation $\text{sqf}(6) = 2 \cdot 3$ implies on its own, and the signature records whether $3 \mid p-1$ without recording the resulting near-determinism inside those cells. (The effect is sharpest in the cells where $p-1$ has no small odd prime factor at all: for $v_2(p-1) = 1$, no prime $q \leq 13$ dividing $p-1$ and exactly one larger factor, the 2×2 table for $(2, 6)$ is exactly diagonal over 1,775,686 primes, giving $\phi = +1$. Across the 47 cells with $v_2(p-1) = 1$, $3 \nmid p-1$ and at least 200 primes, the within-cell ϕ ranges from $+0.39$ to $+0.98$, so the near-determinism is a feature of that family of cells rather than of a single one.) The triple-completed-group mean absolute residual of 0.0179 is substantially driven by this one pair: excluding it, the figure is 0.0077, still eightfold the value for the remaining pairs of 0.0010.

Conditioning additionally on $(\chi_a(p), \chi_b(p))$ reduces (2, 6) to +0.0055, in line with every other pair — so the outlier is explained by the entanglement account rather than in tension with it. Note that (2, 10) and (5, 15) are more negative *after* conditioning than before, and that (5, 10), (5, 15), (6, 10), (10, 15) and (7, 21) change sign under conditioning: the common-cause channel had been masking an underlying repulsion.

Second, additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ removes the negative residuals. We put it this way deliberately: the signed group means do not both fall to zero. The triple-completed mean stays at +0.0049 while the mean for the remaining pairs *rises* from +0.0010 to +0.0052, and the two averages are taken over different eligible populations (about 73% of primes for signature conditioning against 23% here), so they are not components of a single decomposition. Every one of the six negative values above becomes positive and small (+0.0031, +0.0066, +0.0045, +0.0065, +0.0040, +0.0048 respectively), and the triple-completed and remaining groups become indistinguishable at +0.0049 and +0.0052 (Table 7). That common small positive floor of about +0.005 is present for all 66 pairs and is plausibly attributable to residual odd-prime conditions shared through $p - 1$ beyond what our signature records. We note that a *quartic* channel is not available as an explanation here: a fourth power is a square, and in the only quadrant contributing defined correlations both bases are quadratic nonresidues, so no further 2-power residue condition can bind.

Conditioning on the quadratic characters is precisely what removes character-level entanglement, and that it flattens the triple-completed pairs is consistent with the residual as the statistical trace of the relation $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$ — the same object that makes Theorem 6 true.

One part of this can be made exact, though it is a statement about support and not about the residual's size. Theorem 6 asserts that the joint Artin event is empty on the stratum $\chi_a(p) \neq \chi_b(p)$, and this is what the contingency tables show: summing the joint-Artin cell over all signature classes and all 66 pairs, the count of primes $p \leq 10^9$ that are Artin for both a and b with $\chi_a(p) \neq \chi_b(p)$ is

$$0 \quad (\text{out of } 50,847,524 \text{ primes, for every one of the 66 pairs}),$$

while on the complementary stratum the joint density is 0.299. The support restriction is therefore a theorem rather than a conjecture. It does not by itself determine the sign or the magnitude of any residual: a support constraint of this kind is compatible with zero covariance, as the trivial example of two independent fair signs with Artin indicators given by their negative-sign indicators shows. What the residual measures is a quadratic-information component; that this component is the pointwise law's statistical trace is an interpretation the data support, not a consequence of the theorem.

One caveat on the strength of the flattening step, in the other direction. Within a (χ_a, χ_b) stratum the exclusion cannot operate at all: by criterion (1) an Artin prime for a has $\chi_a(p) = -1$, so all the *joint* Artin mass sits in the single quadrant $(-1, -1)$. In each of the other three quadrants at least one of the two indicators is identically zero, so the within-cell correlation is undefined and the conditional covariance vanishes; this is not the same as saying neither indicator varies there. Conditioning on (χ_a, χ_b) therefore *restricts attention to the stratum on which the constraint of Theorem 6 is vacuous*, and the vanishing of the residual there is in part a structural consequence of that restriction rather than an independent confirmation. What the step establishes is the localisation: the residual lives in the character channel and not elsewhere in the signature. We state it in that weaker form.

The picture that emerges is cleanly layered:

$$\underbrace{\text{common cause } p-1}_{\approx 95\%} + \underbrace{\text{quadratic entanglement } \chi_a \chi_b = \chi_{\text{sqf}(ab)}}_{\text{residual, localised on triple-completed pairs}} + \underbrace{\text{higher-order sharing}}_{\text{small positive floor}}.$$

14. COMPARISON WITH THE PREDICTED JOINT DENSITIES

The measurements of Sections 11–13 identify the mechanisms empirically. We now check them against theory quantitatively, by computing the joint densities predicted by the Matthews–Moree–Stevenhagen framework and comparing with the observed counts pair by pair.

Write $d_a = \text{sqf}(a)$. The heuristic, in the form originating with [24] and systematised by the character-sum method of [23] (with the general framework in [20]), is an inclusion–exclusion over the degrees of the Kummer extensions that detect the conditions “ a is an ℓ -th power residue”:

$$\mathbf{P}(p \in \mathcal{A}_a \cap \mathcal{A}_b) = \sum_{m,n \text{ squarefree}} \frac{\mu(m)\mu(n)}{[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}]}, \quad L = \text{lcm}(m, n), \quad (3)$$

and the entanglement enters through the degree in the denominator. Generically

$$[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}] = \varphi(L) m n,$$

but the degree drops by the order of the subgroup of quadratic classes realised inside $\mathbb{Q}(\zeta_L)$, taken from the group

$$V(m, n) = \langle d_a \text{ (if } 2 \mid m), d_b \text{ (if } 2 \mid n) \rangle \subseteq \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$$

whose quadratic field lies inside $\mathbb{Q}(\zeta_L)$, i.e. whose conductor divides L . The point of contact with Theorem 6 is that when both $2 \mid m$ and $2 \mid n$ the group V contains not only d_a and d_b but their product $d_a d_b \sim \text{sqf}(ab)$, so the barring character $\chi_{\text{sqf}(ab)}$ of Theorem 6 is present in the model as a potential source of degree collapse. Two things must be said precisely here, because a natural stronger reading is false. First, the product class lies in $V(m, n)$ for *every* pair of bases whenever both indices are even; its membership is automatic and has nothing to do with whether a third base of our set completes a multiplicative triple. Second, membership in V is not collapse: a class contributes only when its conductor divides L . These two conditions do not coincide, in either direction. For (7, 11) — a pair outside every triple of our set — the product class 77 has conductor 77, which divides $L = 154$, while the individual conductors 28 and 44 do not, so the product class does contribute a collapse there. Conversely for (2, 6) — a pair inside the triple (2, 3, 6) — the nontrivial classes of V have conductors 8, 24 and 12, all divisible by 4, and no squarefree L is, so no term of (3) receives this correction at all. The triple-completion classification used in our tables is therefore a property of the chosen base set, not a field-theoretic invariant of the pair, and it is not equivalent to the occurrence of an extra degree collapse. We evaluate (3) by summing the ramified part exactly over all squarefree m, n supported on $S = \{2\} \cup \{q : q \mid d_a d_b\}$ and folding the rest into the Euler product $\prod_{q \notin S} (1 - \frac{2q-1}{q^2(q-1)})$, truncated at 3×10^6 .

As a check on the implementation, the same code specialised to a single base reproduces the classical Hooley densities: 0.373956 for bases with $d \not\equiv 1 \pmod{4}$ (Artin’s constant), 0.393638 for $d = 5$, 0.376368 for $d = 13$, against measured 0.373957, 0.393642, 0.376362.

Observation 14 (The model reproduces the measurements). Over all 66 pairs at $x = 10^9$, the mean absolute relative error between the predicted and measured joint densities is 0.012%, with maximum 0.040% (attained at (21, 29)). In particular the model predicts the sign reversal: for the negatively correlated pairs (2, 10) and (3, 15) it gives $j = 0.132776$ and $\phi = -0.0302$ for both, against measured $j = 0.132776, 0.132769$ and $\phi = -0.0302, -0.0303$ respectively.

This is the quantitative counterpart of Sections 11 and 13. It is a comparison of unconditional joint densities, so it does not by itself validate any conditional mechanism, but it is a demanding test and the framework passes it. The 66-pair pattern — positive correlation almost everywhere,

TABLE 8. Predicted versus measured joint Artin densities, selected pairs at $x = 10^9$. “TC” marks triple-completed pairs. The final two rows are the two negatively correlated pairs.

a	b	measured j	model j	rel. err.	measured ϕ	model ϕ	TC
5	13	0.163939	0.163943	+0.00%	+0.0667	+0.0667	
5	10	0.161962	0.161922	−0.02%	+0.0624	+0.0623	✓
5	15	0.161934	0.161922	−0.01%	+0.0622	+0.0623	✓
3	7	0.149965	0.149971	+0.00%	+0.0432	+0.0433	✓
13	17	0.150249	0.150261	+0.01%	+0.0383	+0.0383	
2	3	0.147361	0.147349	−0.01%	+0.0321	+0.0321	✓
2	6	0.147334	0.147349	+0.01%	+0.0320	+0.0321	✓
7	21	0.144754	0.144728	−0.02%	+0.0238	+0.0238	✓
15	21	0.144725	0.144728	+0.00%	+0.0237	+0.0238	
3	15	0.132769	0.132776	+0.01%	−0.0303	−0.0302	✓
2	10	0.132776	0.132776	+0.00%	−0.0302	−0.0302	✓

negative precisely on the two pairs where it is observed to be negative, with magnitudes ordered as observed — is not merely *consistent with* the Matthews–Moree–Stevenhagen framework; it is reproduced by it to within a few parts in 10^4 , with no fitted parameters. Correspondingly, the same quadratic class $\text{sqf}(ab)$ governs the deterministic law of Theorem 6 and enters the heuristic through the group $V(m, n)$. We stop short of calling these one object: the class lies in $V(m, n)$ for every pair once both indices are even, while it lowers a degree only when its conductor divides L , and the two conditions come apart in both directions as noted in Section 14.

Remark 15. The residual discrepancies are uniformly small (all 66 pairs below 0.040%) and we have not pursued them; ordinary Hooley-type errors of this size are expected at $x = 10^9$, since the convergence in Hooley’s theorem is only logarithmic, and our Euler products are truncated at 3×10^6 . We record one methodological warning, since it cost us a draft. Computing $\varepsilon(m, n)$ as 2 raised to the number of classes of $V(m, n)$ with conductor dividing L — rather than as the order of the subgroup of $V(m, n)$ realised in $\mathbb{Q}(\zeta_L)$ — inflates ε to 8 on exactly those pairs with $d_a \equiv d_b \equiv 1 \pmod{4}$, where all three of $d_a, d_b, \text{sqf}(ab)$ have odd conductor. Since $V(m, n)$ has two generators, $|V(m, n)| \leq 4$ and $\varepsilon = 8$ is impossible. The error is invisible in the single-base specialisation, which is why the check above does not catch it; it inflates the mean relative error over the 66 pairs from 0.012% to 0.037% and concentrates the damage on the ten pairs drawn from $\{5, 13, 17, 21, 29\}$.

15. COMPUTATIONAL CONTENT: A MEASURED NEGATIVE

The correlations above are large by the usual descriptive measures: $\delta = -0.01414$ with a Pearson statistic of 10,167 on one degree of freedom, and individual cross-base values of ϕ/σ_0 up to 476. Those statistics are descriptive, not calibrated: they describe 5×10^7 enumerated primes rather than a large effect. The question of this section is whether the correlation is *computationally* worth anything — could it predict generator availability, or yield a faster algorithm? We answer by measurement, for the estimators and the search workload described below, and within that scope we find no computational use for it. We record it because the negative is informative and because the protocol transfers to any claimed predictive advantage in this area.

conditioning	entropy (nats)	gain vs marginal	incremental
marginal	0.661047	—	—
joint residues mod 120	0.241529	0.419517 (0.605 bits)	—
+ predecessor Artin status	0.241520	—	0.000009
joint residues mod 840	0.237235	0.423812	—
+ predecessor Artin status	0.237217	—	0.000018

TABLE 9. *Empirical conditional-entropy diagnostic* over 50,847,530 pairs below 10^9 (resubstitution: each cell’s probability is estimated from the same pairs it scores). The elementary joint-residue baseline carries 0.42 nats; the correlation adds of order 10^{-5} nats *in this descriptive sense*. This is not a held-out measurement.

baseline	estimator	gain, even→odd	gain, odd→even
joint residues mod 120	A	-3.2×10^{-6}	-4.9×10^{-6}
	B	$+1.9 \times 10^{-6}$	$+9.4 \times 10^{-8}$
joint residues mod 840	A	-6.11×10^{-5}	-6.08×10^{-5}
	B	-2.55×10^{-5}	-2.71×10^{-5}

TABLE 10. *Held-out* gain, in nats per test pair, from adding the predecessor’s Artin status to the joint-residue cells. Even→odd: probabilities are fitted on the pairs whose successor prime has least significant hash bit 0 and scored on the complementary half (25,421,106 pairs); odd→even is the reverse (25,426,424 pairs). Jeffreys ($+\frac{1}{2}$) smoothing is applied in every cell. Estimator A splits every cell on the predecessor’s status; estimator B does not split a cell whose successor status is constant in the fit half, avoiding a smoothing penalty when that constancy persists. A hash split is used rather than a range split because the Artin density drifts with x . At modulus 120 the sign depends on the estimator; at modulus 840 both estimators lose.

The right baseline, and two different quantities. A correlation can look large against a naive reference and carry no usable information once the elementary structure is taken into account. For consecutive primes that structure is the *joint residue pair*: a practitioner asking whether a is a primitive root mod p already knows $p \bmod 4 \operatorname{sqf}(a)$, and the Legendre symbol $\left(\frac{a}{p}\right)$ is a function of it. The correct baseline for a predictive claim is therefore not the marginal density of Art but the conditional distribution given the joint residues.

Two quantities must be distinguished, and earlier drafts of this work conflated them.

- An *empirical conditional-entropy diagnostic*: estimate each cell’s conditional probability from the data and score those same data. This measures how much of the variation the conditioning explains, and it is a legitimate descriptive summary — but a finer nested model can never do worse, so it is *not* evidence of predictive skill.
- A *held-out* measurement: fit the cell probabilities on one part of the data and score a disjoint part. Only this can show predictive value.

Table 9 gives the first quantity on all 50,847,530 consecutive pairs below 10^9 , and Table 10 the second.

What the in-sample increment is made of. The in-sample increment of Table 9 is 0.000009 nats at modulus 120 and 0.000018 nats at modulus 840. For scale only: with no association and regular i.i.d. sampling, its expectation is $k/(2N)$, where k counts the cells in which both the predecessor’s and the successor’s status vary ($k = 256$ and $k = 1,599$), that is 2.5×10^{-6} and 1.6×10^{-5} nats. That reference is heuristic — the cells are sparse and consecutive pairs overlap — so we calibrate by simulation instead.

The simulation is a *residue-status null*. Every prime’s Artin status is redrawn independently, with probability equal to the observed Artin rate of its residue class modulo M , and the census tables are rebuilt from the simulated statuses. Statuses are drawn per prime, so each simulated pair is still (status of p_n , status of p_{n+1}) and the overlap between consecutive pairs is preserved; by construction the predecessor’s status carries no information beyond residues modulo M . Over 100 replicates for each M :

- with $M = 120$ the null reproduces the reference scale (mean 2.53×10^{-6} , standard deviation 2.1×10^{-7} , for the mod-120 increment), and no replicate reaches the observed 9.18×10^{-6} ;
- with $M = 840$ the null gives a mod-120 increment of mean 8.82×10^{-6} (standard deviation 4.9×10^{-7}), and 24 of the 100 replicates reach or exceed the observed value.

A model in which statuses depend on nothing but $p \bmod 840$ reproduces the excess over the mod-120 baseline, so reproducing it requires no predecessor-specific information. The residue channel involved is the prime 7 of $p - 1$, which the mod-120 cells do not see: primes $p \equiv 1 \pmod{7}$ are Artin base 10 at rate 0.3284, against 0.3831 for the other classes (ratio 0.857, consistent with the heuristic factor $6/7$ for that channel). At modulus 120 estimator B of Table 10 gains slightly ($+1.9 \times 10^{-6}$ and $+9.4 \times 10^{-8}$). We have not tested that held-out figure against the residue-status null, and in view of the in-sample result we do not treat it as evidence of information specific to the predecessor’s status.

Against the mod-840 baseline the observed increment, 1.83×10^{-5} , lies slightly above the $M = 840$ null (mean 1.71×10^{-5} , standard deviation 6.8×10^{-7} ; 2 of the 100 replicates reach it). We leave this small in-sample excess unresolved: the null is a simulation benchmark with its own assumption (independent statuses given the residue class), 100 replicates resolve tail frequencies only coarsely, and neither held-out estimator gains at modulus 840.

Within the estimators above, then, the predecessor’s Artin status shows no predictive value beyond the joint residues modulo 840, and its apparent value beyond residues modulo 120 is reproduced by a residue-only model. We state this as a model- and workload-limited finding, not as an information-theoretic bound: nothing here bounds the oracle conditional mutual information, other estimators or other ranges. The same trap appears one level coarser, and a reader who repeats this should expect it: conditioning on the gap class $g \bmod 40$ *alone* produces a spurious apparent gain of about 0.02 nats, because $g \bmod 40$ does not determine the individual residues $p_n \bmod 40$ and $p_{n+1} \bmod 40$; the apparent signal is residue information re-appearing. The joint residues, at a modulus that includes every small channel of $p - 1$ in play, are the correct baseline.

What the between-cell share measures. The global anticorrelation admits an exact two-term decomposition. Writing X, Y for the predecessor’s and successor’s Artin indicators and C for the joint-residue cell, the law of total covariance gives

$$\delta = \underbrace{\frac{1}{\text{Var}(X)} \sum_c \pi_c r_c (1 - r_c) \delta_c}_{\text{within}} + \underbrace{\sum_c (w_c^1 - w_c^0) p_c}_{\text{between}}$$

with $\pi_c = P(C=c)$, $r_c = P(X=1 \mid C=c)$, $w_c^x = P(C=c \mid X=x)$, $p_c = P(Y=1 \mid C=c)$ and δ_c the within-cell association. This is an identity, not a model. In a cell where the predecessor's status never varies δ_c is undefined; under this and every convention below such a cell contributes zero within-cell association (equivalently, the missing conditional probability is completed by the observed one). The driver accumulates each within and each between sum independently and asserts the identities; at 10^9 they hold to machine precision. The covariance form places 98.69% (mod 120) and 99.55% (mod 840) of δ between joint-residue cells, with within-cell terms -0.000186 and -0.000064 .

The share depends on the weighting convention. With the Kitagawa ordering and $P(C=c \mid X=1)$ weights, whose between term is $\sum_c (w_c^1 - w_c^0) P(Y=1 \mid X=0, C=c)$, it is 96.14% (mod 120) and 97.81% (mod 840); with $P(C=c \mid X=0)$ weights it is 99.22% and 99.71%; with symmetric weights 97.68% and 98.76%. Across these four conventions the between share lies between 96.1% and 99.2% at modulus 120, and between 97.8% and 99.7% at modulus 840. Cells in which the predecessor's status never varies hold about half of all pairs (a fraction 0.49996 at modulus 120, all of them cells with no Artin predecessor), and by the convention just stated they contribute only to the between term.

Both exclusion laws are computationally redundant. An exclusion law can be mathematically correct and computationally useless, and both of ours are instances of that. The gap law (Theorem 2) is the statement that for $g \equiv 20 \pmod{40}$ the product $\chi_{10}(p)\chi_{10}(p+g)$ is the constant -1 (equivalently $\chi_{10}(p(p+g)) = -1$), so the two characters are opposite and at most one of the two primes can be Artin. The constant sign follows from quadratic reciprocity applied to the shift g , together with multiplicativity; multiplicativity alone does not fix which sign the shifted product takes. But a practitioner need not track predecessor status or gap state to learn this: reading $p \pmod{40}$ gives $\chi_{10}(p)$ directly, and $p+g \pmod{40}$ gives $\chi_{10}(p+g)$. The law is a statement about characters that a character computation already answers.

The same is true of the same-prime law (Theorem 6) and its triple corollary (Theorem 7): they follow from $\chi_c = \chi_a \chi_b$, so if a and b are Artin then both characters are -1 and $\chi_c = +1$, whence c is not Artin whenever the three bases are units modulo p (i.e. $p \nmid abc$). To be precise about the scope of “redundant”: *once the relevant individual quadratic characters are already available*, these statements supply no further rejection beyond what those characters give. Both deductions combine characters that are already in hand; for the gap law, reciprocity is what makes the χ_5 character available in the first place. Neither saves a modular exponentiation that a character-aware implementation was not already saving. We verified the gap law exhaustively at 10^9 — 2,214,511 pairs with $g \equiv 20 \pmod{40}$, of which 984,678 have an Artin predecessor, and *zero* are doubly-Artin — but exhaustive verification of a redundancy is still a redundancy.

The quadratic filter is elementary. The quadratic filter expressed by the exclusion laws is the necessary condition of §3, already applied by a character-aware implementation: a primitive root must be a quadratic non-residue, so $\left(\frac{a}{p}\right) = -1$. It is answerable in $O(1)$ arithmetic by reciprocity rather than by exponentiation, and it is exactly the information that the conductor-40 quadratic channel of §9 carries. That is the honest reading of the decomposition: the paper *explains* why a standard filter works, and identifies which channel carries the effect, but it does not supply a new one.

We measured it in a scanning search for Artin-base-10 primes at $p \approx 10^{15}$, where factorising $p-1$ dominates. The three strategies below are single-threaded loops on a 32-core machine; the recorded times are kernel times for the search loop itself, with the shared small-prime setup (identical for all three, and amortised in any real deployment) excluded.

strategy	time (s)	factorizations of $p - 1$	ratio to naive
naive (factor every prime)	0.1642	722	1.00×
non-residue filter first	0.0841	376	1.95×
+ small- ℓ early exit	0.0732	300	2.24×

TABLE 11. Finding 300 Artin-base-10 primes above 10^{15} ; 722 primes visited. The small- ℓ early exit tests the channels $\ell \in \{3, 5, 7, 11, 13, 17, 19\}$ of $p - 1$ before the full factorisation, rejecting on first failure (402 cheap exponentiations). The last step is about 15% faster, equivalently about 13% less elapsed time, than the non-residue filter alone; the exact counters (primes, factorizations, exponentiations) reproduce independently, while timings are single-run and hardware-dependent.

Ordering the small prime channels of $p - 1$ before the full factorisation — testing $a^{(p-1)/\ell} \not\equiv 1$ for small $\ell \mid p - 1$ and rejecting on the first failure — saves 20% of the factorizations. This is a *test-ordering* principle rather than a new algorithm, and it is workload-dependent in a way worth stating: if the task is to find *any* primitive root for a *given* p , then $p - 1$ is factored once and cached, and the early exit saves the cheap exponentiations but not the factorisation at all. The gain exists only for scanning searches over fresh primes.

What would change the verdict. This is a negative assessment of the evidence, not an impossibility theorem. The conditioning above does not prove exact conditional independence, and the verdict would change if: (i) a repeated workload existed in which many compatible primes or candidate bases are tested and the required labels are already available; (ii) a new *deterministic* exclusion were found that rules out a case the character deductions do not already rule out; (iii) an actionable decision followed from the correlation — skipping a soundly excluded candidate, or reordering costly tests — with net cost falling after factorisation, feature acquisition and bookkeeping; and (iv) correctness were preserved, since a predictor may order tests but cannot replace certification. None of these holds in the workloads tested here. We have not evaluated a cross-base predictive workload at all, so the negative is demonstrated for the consecutive-prime axis and for the scanning-search workload; the same-prime axis is covered only by the redundancy argument above. The results are best understood as describing arithmetic structure, and as identifying exactly which elementary structure the observed correlations are made of.

16. DISCUSSION

What is new here. The individual ingredients are classical: quadratic reciprocity, the quadratic-residue obstruction to primitive roots, the consecutive-prime bias of [14], and the Kummer-degree density computations of [24, 23]. The contributions of this paper are: (i) the observation that these combine into deterministic exclusion laws on both axes — a gap law for consecutive primes at base 10 (Theorem 2) and a same-prime pair law with a triple corollary (Theorem 6, Theorem 7). The gap law is the base-10 instance of the phenomenon identified by Tinková–Waxman–Zindulka [13], and we claim no priority for it; (ii) a measurement, over the primes below 10^9 , of the consecutive-prime Artin correlation, its gap profile, and the ω repulsion, together with the cross-base correlations $\phi(a, b)$ and their joint-density comparison; (iii) a descriptive reduction showing that, depending on the weighting convention, 96.1–99.2% (modulus 120) and 97.8–99.7% (modulus 840) of the consecutive-prime anticorrelation lies

between joint-residue cells, and that 95% of the mean cross-base correlation is removed by conditioning on the fine signature of $p - 1$ — that is, both correlations are largely accounted for by the same elementary structure — the residues of p and the factorisation of $p - 1$ — rather than by an interaction (reductions of different estimands on different selected populations, not a complete decomposition); and (iv) a measured audit of the computational content of those correlations, which finds no computational use for them within the estimators and the workload tested (§15). The exclusion laws are immediate applications of standard quadratic reciprocity; their simplicity is a feature, not a limitation, and the same is true of the audit.

Relation to runs of Artin primes. Pollack [17] proved (under GRH) that for every nonsquare $g \neq -1$ and every m there are infinitely many runs of m consecutive primes each having g as a primitive root. Baker and Pollack [2] proved unconditionally that if Q contains at least $\exp(Cm)$ multiplicatively independent primes, then infinitely many runs of m consecutive primes each have some element of Q as a primitive root, with the primes lying in a bounded-length interval. Our results add an arithmetic constraint to the base-specific version: for base 10, no run of all-Artin consecutive primes can cross a gap $\equiv 20 \pmod{40}$. We have not inferred longer-run frequencies from the pair statistic, and we emphasise that the existence results above concern rare favourable configurations rather than the typical behaviour measured here.

Relation to the Kummer-degree theory. The pointwise obstruction of Theorem 6 and the Kummer-degree heuristic of §14 are both governed by the same quadratic class $\text{sqf}(ab)$: in the first it appears as a congruence barring half the primes, in the second as a member of the group of quadratic classes that can reduce a degree. We do *not* claim these are the same phenomenon. The product class belongs to that group for every pair, whereas it lowers a degree only when its conductor divides the relevant cyclotomic level, and §14 exhibits pairs separating the two. The agreement of the Matthews–Moree–Stevenhagen densities with measurement to 0.012% mean relative error is a check of the density theory at this scale, not a new derivation of it.

Future work. Four extensions suggest themselves. (1) *Other bases*: the same pipeline with $a = 2, 3, 5, 6, 7$ would test the conductor prediction after Corollary 3 (conductor 8 for $a = 2$, so exclusion classes mod 8; conductor 24 for $a = 6$). (2) *Scale*: extending the consecutive census to 10^{12} would help distinguish decay from persistence in $\delta(x)$. (3) *Theory*: a derivation of $\delta(g)$ from the singular series of [9] combined with Hooley’s method, at least for the dominant channels, would replace empirical conditioning with a formula. (4) *Audit protocol*: the measurement in §15 is a general recipe — compare a claimed predictive advantage against the elementary necessary-condition baseline, using a proper scoring rule, and report end-to-end cost including feature acquisition — and it would be worth applying to other claims of algorithmic value from arithmetic correlations.

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DECLARATION OF INTEREST

No potential conflict of interest was reported by the author.

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DATA AVAILABILITY

All code required to reproduce every number in this paper is supplied in the accompanying reproduction package: the C segmented sieve and census programs, the independent recomputation, the analysis scripts, the verification driver for the tables of §15, and the Lean development for the exclusion laws. The code, manuscript, results and Lean development are at <https://github.com/jpbald93/artin-correlations>. The per-prime dataset itself (1.8 GB CSV covering all 50,847,531 primes from 7 to 999,999,937: gap, $\omega(p-1)$, Artin indicator, and small residues) is not deposited because of its size, but it is exactly and deterministically reproducible from the included C source on a single machine, and is available from the author on request.

17. OPEN QUESTIONS

The Hardy–Littlewood-based framework of [14] makes the residue transition frequencies of consecutive primes quantitatively predictable, and every channel in our conditioning is built from such frequencies. This motivates the following open questions.

Conjecture 16 (Persistence). *For primes up to x , the consecutive-pair Artin correlation $\delta(x) = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$ is negative for all sufficiently large x .*

We have measured δ at four decade-spaced cutoffs:

$$\delta(10^8) = -0.013363, \quad \delta(10^9) = -0.014140, \quad \delta(10^{10}) = -0.014723, \quad \delta(10^{11}) = -0.014998,$$

the last two from an independent recomputation described in Section 6 (over 455,052,508 and 4,118,054,810 primes respectively). At these four cutoffs the magnitude of δ increases, and the increments -0.000777 , -0.000583 , -0.000275 decrease. No turnover is observed over the measured range.

We draw no asymptotic conclusion from this. Four nested cumulative cutoffs are not four independent observations, and they determine neither an eventual sign nor a limit nor a decay rate. Two specific negative remarks are all that the values support: the ratio of successive increments is not constant (0.75 then 0.47), so a geometric increment model does not describe them — extrapolating geometrically from the first three values predicts $\delta(10^{11}) \approx -0.01516$ against a measured -0.014998 , a discrepancy of 1.6×10^{-4} — and $\delta(x) \log x$ is not constant (-0.246 , -0.293 , -0.339 , -0.380), so a bare $c/\log x$ law does not describe them either.

It is worth recording explicitly that monotone strengthening with decelerating increments does *not* constrain the eventual rate, and in particular does not exclude decay to zero. Writing $t = \log_{10} x$, the function

$$f(t) = -\frac{0.246399}{t} - \frac{1.557171}{t^2} + \frac{41.902805}{t^3} - \frac{164.142380}{t^4}$$

reproduces all four measured values at $t = 8, 9, 10, 11$, is negative for all $t \geq 8$, strengthens across the measured range, and yet satisfies $f(t) \sim -0.246399/t$, so that $f \rightarrow 0$; its turning point lies at $t \approx 11.56$, just beyond the last cutoff. This is an interpolation constructed to make

the point, not a proposed model, and we attach no significance to its coefficients. It shows only that our data are equally compatible with eventual decay to zero and with persistence of a nonzero correlation, and that the question is not settled by extending the census a decade or two further.

We also note that the conjecture as stated — that δ is eventually negative — is compatible with $\delta(x) \rightarrow 0$; eventual negativity and decay to zero are not competing alternatives. A separate reason for caution against assuming a nonzero limit arises in a restricted setting. Consider a *separable* fixed-modulus channel model: write R, S for the residue classes of (p_n, p_{n+1}) modulo a fixed M , and suppose the modelled indicators satisfy $\mathbb{E}[X \mid R, S] = \alpha(R)$, $\mathbb{E}[Y \mid R, S] = \beta(S)$ and $\text{Cov}(X, Y \mid R, S) = 0$, so that the only modelled dependence between the two endpoints runs through the joint law of (R, S) . If that joint law tends to the product of its marginals, as the leading term of the Lemke Oliver–Soundararajan prediction gives for fixed M [14], then $\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] \rightarrow 0$, and hence the modelled $\delta \rightarrow 0$ whenever the limiting source marginal is nondegenerate. Non-uniform singular-series weights across gap classes do not by themselves defeat this conclusion: the gaps grow with x , and it is the surviving total covariance, rather than the non-uniformity of the weights, that would have to be estimated.

The separability hypothesis is essential and is not automatic. Conditioning on the joint pair (R, S) alone does not force independence: taking R, S independent and uniform on the sixteen reduced classes modulo 40 and setting $X = Y = Q(R)Q(S)$, where Q indicates the eight classes on which the relevant quadratic symbol is -1 , gives perfectly equidistributed residue pairs together with $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[XY] = \frac{1}{4}$ and $\text{Cov}(X, Y) = \frac{3}{16}$. So the observation above constrains separable finite-modulus models, not every model built on a fixed modulus. Extending a finite-obstruction approach to a statement about the limit would in addition require joint-distribution information at each finite channel together with a uniform bound on the omitted higher-power-residue obstructions; the quadratic channel visible here is only the first of these. The question of the limiting behaviour is unresolved; finite-range measurements cannot distinguish the two alternatives.

Research question (residue completeness). Does there exist, for every $\varepsilon > 0$, a modulus M such that conditioning on the joint residues of (p_n, p_{n+1}) modulo M reduces some appropriate within-cell dependence measure below ε in absolute value? This is stated as a question rather than a conjecture because a precise formulation requires specifying the order of limits (over x and M), the treatment of sparse and degenerate cells, the dependence criterion (e.g. a covariance decomposition rather than a signed cell mean), and the range of x over which the bound is asserted. We do not know a formulation that avoids these ambiguities.

We remark that a sufficiently uniform Hardy–Littlewood prime k -tuple framework, together with a justified treatment of the no-intermediate-prime restriction, would be the natural setting in which to predict the joint distribution of $(p_n \bmod M, p_{n+1} \bmod M, g)$ via singular series. The pair conjecture alone does not suffice, and the restriction is not an optional refinement: consecutiveness is part of the event, and its indicator

$$\mathbf{1}_{p \text{ prime}} \mathbf{1}_{p+g \text{ prime}} \prod_{1 \leq h < g} (1 - \mathbf{1}_{p+h \text{ prime}})$$

expands into higher-tuple correlations that two-point counts do not determine. This is why [14] proceeds from a k -tuple heuristic rather than from the pair conjecture. Whether such predictions carry the Chebotarev and power-residue information needed for Artin status is a further question again, and requires analysis we do not attempt here.

Conjecture 17 (ω repulsion). $r(\omega(p_n - 1), \omega(p_{n+1} - 1))$ is negative for all large x and tends to 0 as $x \rightarrow \infty$.

The single reported value does not establish eventual sign or limit. We state this as a conjecture consistent with the data.

REFERENCES

- [1] E. Artin, *The Collected Papers of Emil Artin* (S. Lang and J. T. Tate, eds.), Addison–Wesley, Reading, MA, 1965.
- [2] R. C. Baker and P. Pollack, *Bounded gaps between primes with a given primitive root, II*, Forum Math. **28** (2016), no. 4, 675–687.
- [3] J. H. Conway and R. K. Guy, *The Book of Numbers*, Springer-Verlag, New York, 1996.
- [4] P. Erdős, *On the normal number of prime factors of $p - 1$ and some related problems concerning Euler’s φ -function*, Quart. J. Math. Oxford Ser. **6** (1935), 205–213.
- [5] K. (S.) Fan and P. Pollack, *Counting primes with a given primitive root, uniformly*, Mathematika **71** (2025), no. 4, Paper No. e70055.
- [6] S. R. Garcia, E. Kahoro, and F. Luca, *Primitive root bias for twin primes*, Exp. Math. **28** (2019), no. 2, 151–160.
- [7] S. R. Garcia, F. Luca, K. Shi, and G. Udell, *Primitive root bias for twin primes II: Schinzel-type theorems for totient quotients and the sum-of-divisors function*, J. Number Theory **208** (2020), 400–417.
- [8] R. Gupta and M. R. Murty, *A remark on Artin’s conjecture*, Invent. Math. **78** (1984), no. 1, 127–130.
- [9] G. H. Hardy and J. E. Littlewood, *Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.
- [10] D. R. Heath-Brown, *Artin’s conjecture for primitive roots*, Quart. J. Math. Oxford Ser. (2) **37** (1986), no. 145, 27–38.
- [11] C. Hooley, *On Artin’s conjecture*, J. Reine Angew. Math. **225** (1967), 209–220.
- [12] O. Klurman, I. E. Shparlinski, and J. Teräväinen, *On Artin’s conjecture on average and short character sums*, Bull. Lond. Math. Soc. **57** (2025), no. 8, 2429–2443; [arXiv:2412.13355](https://arxiv.org/abs/2412.13355).
- [13] M. Tinková, E. Waxman, and M. Zindulka, *Artin twin primes*, J. Number Theory **245** (2023), 203–232, doi:10.1016/j.jnt.2022.10.006; [arXiv:2010.15988](https://arxiv.org/abs/2010.15988).
- [14] R. J. Lemke Oliver and K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, Proc. Natl. Acad. Sci. USA **113** (2016), no. 31, E4446–E4454.
- [15] P. Moree, *Artin’s primitive root conjecture — a survey*, Integers **12** (2012), no. 6, Art. A13; doi:10.1515/integers-2012-0043.
- [16] A. Perucca and I. E. Shparlinski, *Uniform bounds for the density in Artin’s conjecture on primitive roots*, Bull. Lond. Math. Soc. **57** (2025), 978–991.
- [17] P. Pollack, *Bounded gaps between primes with a given primitive root*, Algebra Number Theory **8** (2014), no. 7, 1769–1786.
- [18] J. Bald, *Quadratic exclusion laws for consecutive Artin primes in arbitrary bases*, unrefered preprint (2026), Zenodo, doi:10.5281/zenodo.22865343; code and data at <https://github.com/jpbald93/quadratic-exclusion-laws>.
- [19] L. Goldmakher, G. Martin, and P. Péringuey, *Refinements of Artin’s primitive root conjecture*, preprint (2025), [arXiv:2502.19601](https://arxiv.org/abs/2502.19601).
- [20] O. Järvinen, A. Perucca, and P. Sgobba, *Unified treatment of Artin-type problems II*, Res. Number Theory **11** (2025), Art. 42, 19 pp.
- [21] N. Kimmel, *Artin’s primitive root conjecture in number fields and for matrices*, Acta Arith. **216** (2024), no. 4, 329–348.
- [22] H. W. Lenstra, Jr., *On Artin’s conjecture and Euclid’s algorithm in global fields*, Invent. Math. **42** (1977), 201–224.
- [23] P. Moree and P. Stevenhagen, *Computing higher rank primitive root densities*, Acta Arith. **163** (2014), no. 1, 15–32.
- [24] K. R. Matthews, *A generalisation of Artin’s conjecture for primitive roots*, Acta Arith. **29** (1976), no. 2, 113–146.
- [25] P. Sgobba, *Unconditional results for Artin-type problems over number fields*, preprint (2025), [arXiv:2508.08996](https://arxiv.org/abs/2508.08996).

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